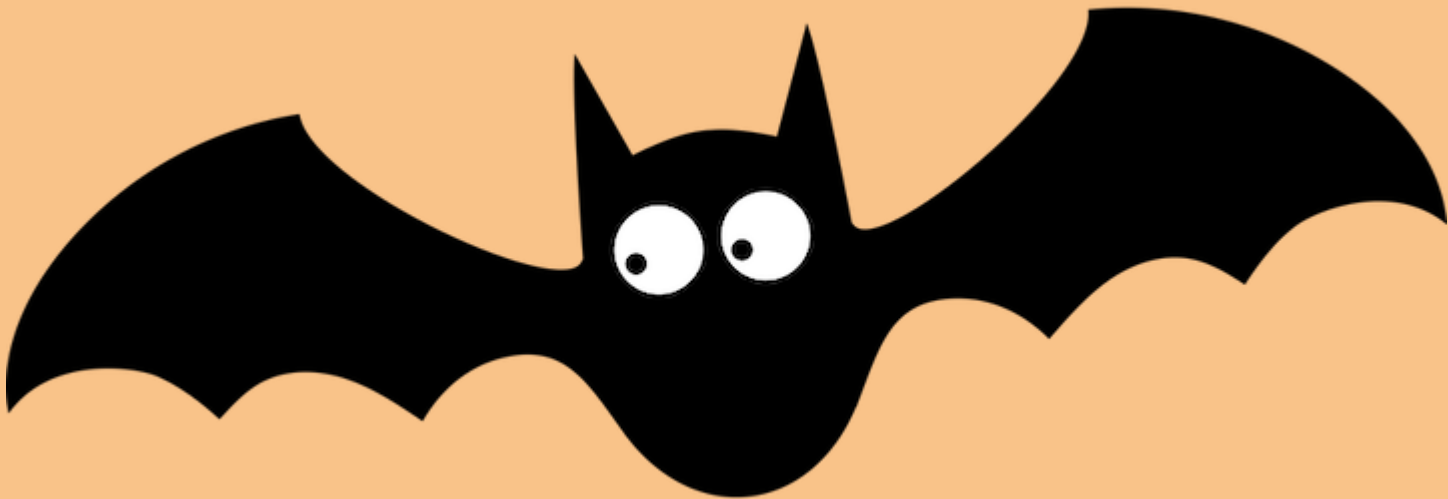




Gradient Boosting

Son Nguyen

Gradient boosting

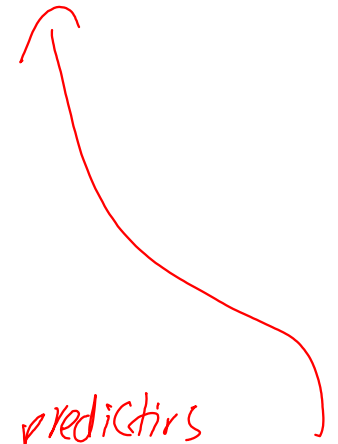
- ① The idea
- ② How Stumps work for regression
- ③ R^2 / MSE
- ④ Learning rate / # Stumps
- ⑤ SAS → Assignment 5
- ⑥ Python dog / cat recognition

Gradient Boosting

We use a regression problem to demonstrate
Gradient Boosting

salary y	stump 1 prediction y_1	error ($y - y_1$)
80	80	0
70	60	10
90	100	-10
100	60	40
50	70	-20
30	80	-50
120	200	-80
200	300	-100

the error of
Stump 2



Round 1

next round

Instead of predictors
 y , the stump will predict

X	Y ← target
30	80
40	90
60	60
50	70

original data

Build
→
a model
↑
(weak)

X	Y	$\hat{y}_1$ Model 1	error $(Y - \hat{y}_1)$
30	80	80	0
40	90	120	-30
60	60	40	20
50	70	100	-30

ϵ_1 the target
of the 3rd

New data ← target

X	ϵ_1
30	0
40	-30
60	20
50	-30

Build
→
a model

X	ϵ_1	Model 2	Error	model
30	0	10	-10	
40	-30	-20	-10	
60	20	20	0	
50	-30	-40	-10	

data of the second
round

...

original data

X	Y $\leftarrow$ target
30	80
40	90
60	60
50	70

Build
 $\rightarrow$
 a model
 $\uparrow$
 (weak)

X	Y	γ_1 Model 1	shrink γ_1	ϵ_1
30	80	80	8	72
40	90	120	12	78
60	60	40	4	56
50	70	160	16	60

$2 = .1$

original data

New data

X	ϵ_1 $\leftarrow$ target
30	72
40	78
60	56
50	60

Build
 $\rightarrow$
 a model

X	ϵ_1	Model 2 γ_2	shrink γ_2
30	72	70	7
40	78	90	9
60	56	50	5
50	60	40	4

$\leftarrow$ the target
 of the 3rd

data of the second
 round

① You need to decide how many models (rounds)
your gradient boosting model has

② Learning Rate or shrinkage ←

③ Combine all the model to the final model :

$$\text{Final prediction} = \underset{\substack{\uparrow \\ \text{learning rate}}}{\lambda} \begin{pmatrix} \text{prediction} \\ \text{of} \\ \text{model 1} \end{pmatrix} + \lambda \begin{pmatrix} \text{prediction} \\ \text{of} \\ \text{model 2} \end{pmatrix} + \dots$$

Learning rate

- too large (> 1) $\rightarrow$ model perform very bad
- too small ($.0001$) $\rightarrow$ take longer to improve the model

usually learning rate $\approx .01$

MSE = Mean Squared Error

$$R^2 = 1 - \frac{\text{MSE of the model}}{\text{MSE of the } \bar{y} \text{ model}}$$

$$R^2 \leq 1$$

R^2 could be negative

Notice : For linear model : $0 \leq R^2 \leq 1$

What is the $\bar{y}$ model? It is the model that predicts y by $\bar{y}$. This model does not take any input variables.

x	y	$\bar{y}$
30	80	75
40	90	75
50	60	75
60	70	75

$R^2 = 0$; the model is as good as the $\bar{y}$ model

$R^2 < 0$ _____ worse than _____

$R^2 > 0$ _____ better than _____

